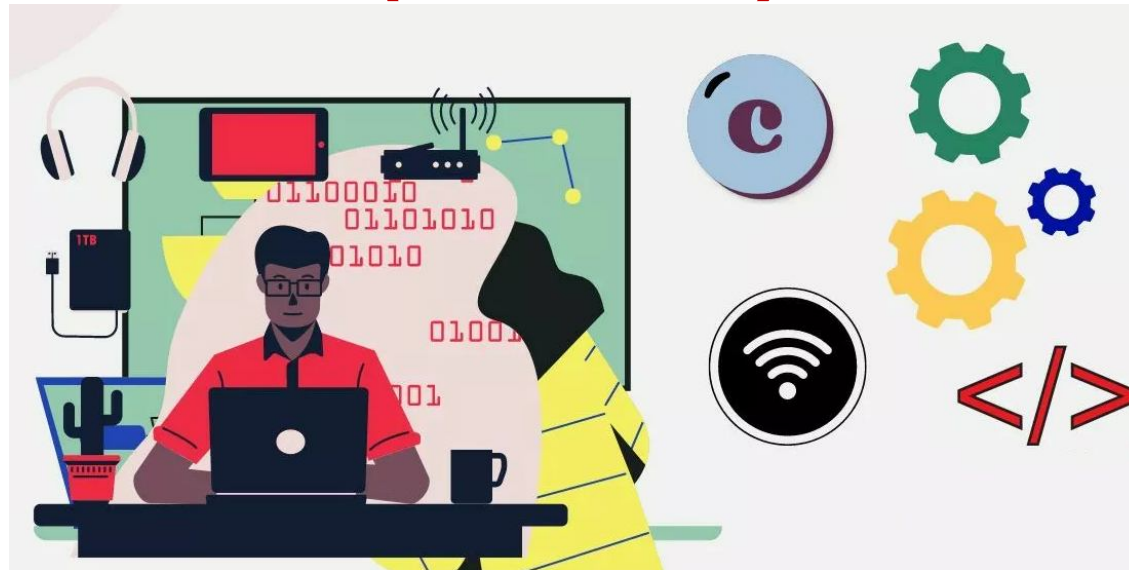


Structured Programming Concepts

(26CSE0105)



Basics of C Programs

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Topics to be covered

- ***Introduction***
- ***Tokens***
- ***Keywords***
- ***Identifiers***
- ***Variables and Constants***
- ***Data Types***

Programming & Coding ???



- **Programming** is the **mental process** of thinking up instructions to give to a machine (like a computer).
- **Coding** is the **process of transforming** those ideas into a written language that a computer can understand.



- **Programming** is a **collaboration** between **humans** and **computers**.

How Programming works?



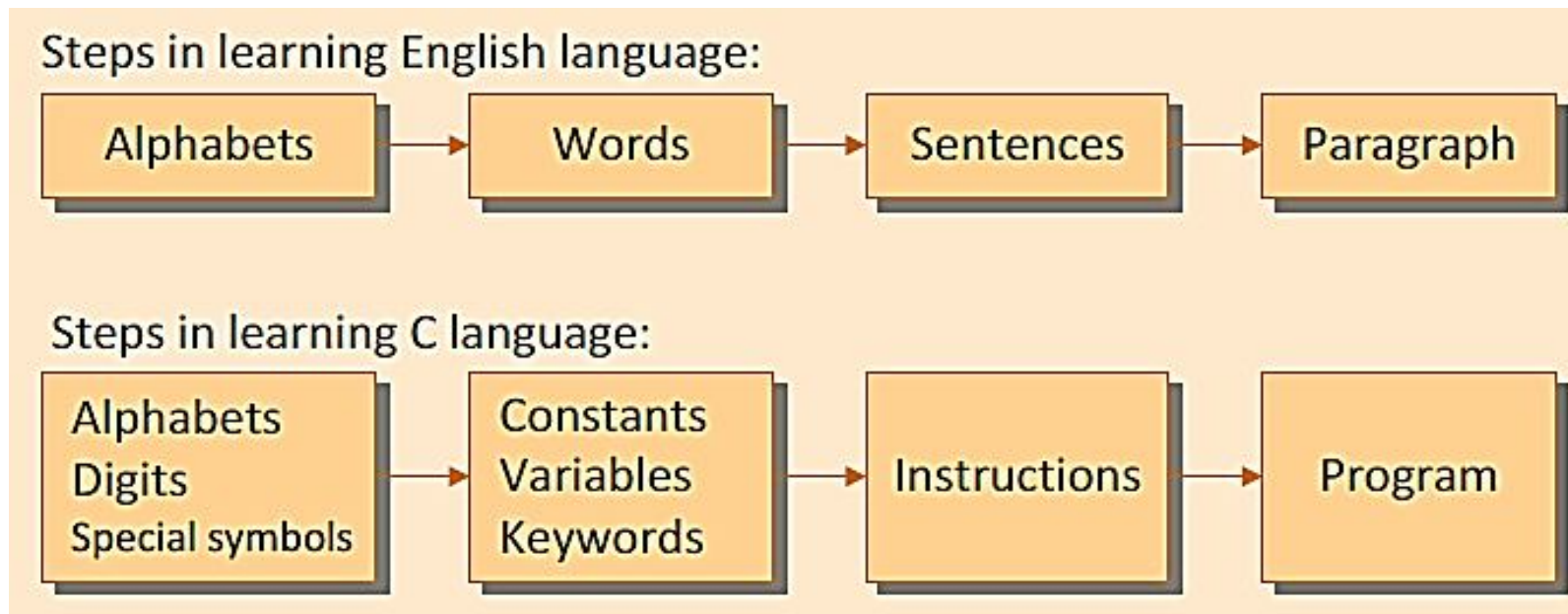
- First, a **programmer** writes code:
 - set of **letters**, **numbers**, and **characters**.
- Next, a **compiler** converts each **line of code** into **machine language**.
- Then, the **computer system** scans the **code** and **executes** it, thereby **performing a task** or **series of tasks**.



Getting Started with C



- *Communicating with a computer* involves *using the language the computer understands*.
- However, there is a *close analogy* between *learning English language* and *learning C language*.



C Character Set

- The **character set** is the fundamental **raw-material** for any language.
- Like natural languages, **computer languages** will also have well **defined character-set**, which is useful to build the programs.
- The **C language** consists of **two character sets**;
 - **Source character set** is useful to construct the statements in the **source program**.
 - **Execution character set** is employed at the time of **execution of program**.

C Character Set

- **Source character set** includes three types of sets;

Alphabets	A, B,, Y, Z a, b,, y, z
Digits	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Special symbols	~ ' ! @ # % ^ & * () _ - + = \ { } [] : ; " ' < > , . ? / \$

C Character Set

- **Execution character set** are also called as **non-graphic characters** because these are invisible and **cannot be displayed** directly.
- Will have effect **only when the program being executed**.

Execution character	Meaning	Result at the time of execution
\n	End of a line	Transfers the active position of cursor to the initial position of next line
\0 (zero)	End of string	Null
\t	Horizontal Tab	Transfers the active position of cursor to the next Horizontal Tab
\v	Vertical Tab	Transfers the active position of cursor to the next Vertical Tab
\f	Form feed	Transfers the active position of cursor to the next logical page
\r	Carriage return	Transfers the active position of cursor to the initial position of current line

Input / Output function

- To **display output** on the screen, we use the **printf()** function.
- We have to use **#include<stdio.h>** at the **starting of the program**.
- The general form of **printf()** function is;

printf ("<format string>", <list of variables>);

- **<format string>** can contain;

%f for printing **real/ float** values

%d for printing **integer** values

%c for printing **character** values

Input / Output function

To make the **program general**, the program itself **should ask** the user to **supply the values** through the **keyboard** during **execution**.

- It can be achieved using a **function** called **scanf()**.
- Use of **&** before the variables in **scanf()** is a must.
- **&** is an '**Address of**' operator, gives the **location number** (**address**) used by the variable in memory.

For example:

```
scanf ( "%d %d %f", &p, &n, &r );
```

Constants, Variables and Keywords

The **combination** of alphabets, digits and special symbols form;
constants, variables and **keywords**.

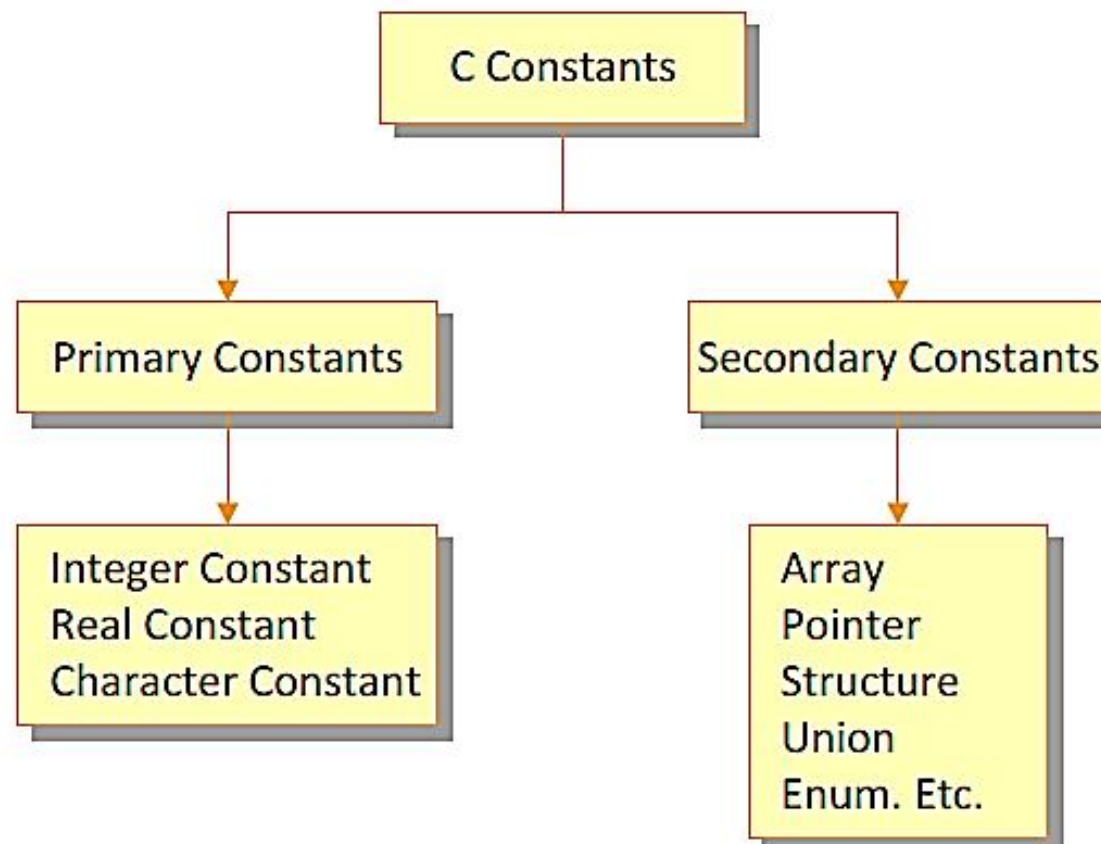
- ❖ A **constant** is an entity that **doesn't change**.
 - ❖ Whereas, a **variable** is an entity that **may change**.
 - ❖ A **keyword** is a word that carries **special meaning**.
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- In any **C program** we typically do **lots of calculations**.
 - The **results** are stored in computer's memory, **memory cells**.
 - To **retrieve** and **use** these values, the **memory cells** are given names called **variable names**.

Constants in C



In programming languages;

- ❖ **constants** are often called **literals**.
- ❖ whereas, **variables** are called **identifiers**.



Rules for Constructing Integer Constants;

- (a) An **integer constant** must have **at least one digit**.
- (b) It must not have a decimal point.
- (c) It can be either **+ve** or **-ve**.
- (d) If **no sign**, it is assumed to be **+ve**.
- (e) **No commas** or **blanks** are allowed.

The range of an Integer constant depends upon the compiler.

for compilers like **GCC**: **-2147483648 to +2147483647**

for compilers like **Turbo C/ C++**: **-32768 to +32767**

Rules for Constructing Real Constants;

The **real constants** could be written in two forms;

- **Fractional form**
- **Exponential form**

- (a) A **real constant** must have **at least one digit**.
- (b) It must have **a decimal point**.
- (c) It could be either **+ve** or **-ve**.
- (d) Default sign is **+ve**.
- (e) **No commas** or **blanks** are allowed.

Range of real constants expressed in exponential form:

-3.4e38 to 3.4e38.

Rules for Constructing Character Constants;

(a) A character constant is a **single alphabet**, a **single digit** or a **single special symbol** enclosed within **single inverted commas** (' ').

For example;

'A' is a valid character constant.

whereas **'A'** is not valid.

A **particular type of variable** can hold only the **same type of constant**.

- An **integer variable** can hold only an **integer constant**.
- A **real variable** can hold only a **real constant**.
- A **character variable** can hold only a **character constant**.

Rules for Constructing Variable Names

(a) Maximum length of a variable name is upto **31 characters**.

Some **compilers** allow variable names length upto **247 characters**.

(b) First character must be an **alphabet** or **underscore** (_).

(c) No commas or **blanks** are allowed within a variable name.

(d) No special symbol other than an **underscore** is allowed.

Keywords in C

- **Keywords** are the words whose meaning has **already been explained** to the compiler.
- There are only **32 keywords** available in **C language**.

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while

➤ The **keywords cannot be used** as **variable names**.



Thank You!

